

UNIVERSITY OF ZAGREB
FACULTY OF MECHANICAL ENGINEERING AND NAVAL ARCHITECTURE

BACHELOR'S THESIS

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ZAGREB, 2026

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ZAGREB, 2026

I hereby declare that I have written this thesis independently, using the knowledge acquired during my studies and the literature cited in the references.

TODO: Zahvale

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ZAVRŠNI ZADATAK

Student: **Andrija Adamović**

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Naslov rada na
hrvatskom jeziku: **Willemsova fundamentalna lema s primjenama**Naslov rada na
engleskom jeziku: **Willems' Fundamental Lemma with Applications**

Opis zadatka:

Classical approach to simulation and controller design for dynamical systems is a model-based approach. A model of system's dynamics is obtained either directly by applying laws of physics on the considered system or indirectly via system identification. The latter is a data-based approach and the data, which consists of recorded input-output trajectories of the system, is used in a suitably defined optimization problem to obtain a model.

Alternative to model-based approaches is the direct data-based approach where simulations and/or controller synthesis is performed directly from data, completely avoiding a model. Such approach has gained significant attention in the control systems community over the past years. There are several reasons for that and the obvious one is that modern IT technologies enable collecting (sensing/recording) of large data sets in an inexpensive way. From a scientific point of view, the direct data-based approach is attractive as it can offer exact results with robustness guarantees, similar to the classical model-based approach.

One of the key results that has inspired and enabled a set of methods in direct data-based approach is the so-called Willems' fundamental lemma. Loosely speaking, this lemma states that all possible trajectories of a linear time invariant (LTI) dynamical system belong to an image space of a suitable defined data-based matrix. The main goal of this thesis is to summarize dominant state-of-the-art approaches to data-based simulation and controller design and to suitably illustrate them on several examples.

The thesis should contain the following:

- Presentation of the Willems' fundamental lemma together with discussion of its origins and meaning.
- Literature overview of the state-of-the-art results on data-based approaches which are based on the Fundamental lemma.
- On a set of suitably selected and designed examples illustrate some of the data-based approaches for simulation and control of LTI dynamical systems.
- Using suitable simulation tools (e.g., MATLAB/Simulink) to investigate applicability of the Fundamental lemma on nonlinear systems.

The thesis must contain the list of used literature as well as any assistance received.

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LIST OF SYMBOLS

Symbol	Description	Unit
x	Example variable	m
t	Time	s

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial intelligence
API	Application programming interface

SAŽETAK

Ovdje upišite sažetak rada na hrvatskom jeziku.

KLJUČNE RIJEČI: ključna riječ jedan; ključna riječ dva

ABSTRACT

Write the thesis abstract in English here.

Keywords: keyword one; keyword two; keyword three

1 Introduction

Classical control systems almost always use a system model to synthesise a control algorithm.

1.1 Motivation

Describe the background and motivation for the thesis.

1.2 Literature Overview

2 The Fundamental Lemma

2.1 Theoretical Background

The goal of this section is to provide introduction to mathematical concepts needed for understanding the rest of the thesis.

2.1.1 LTI Dynamical Systems

Although Willems' Fundamental Lemma stems from the behavioural approach to systems theory, in this thesis it is formulated using the classical state-space LTI system formulation. This formulation is simpler, yet sufficient for the further chapters of the thesis.

Definition 1. A discrete-time linear time-invariant (LTI) system is a dynamical system whose dynamics are described by:

$$x(t+1) = Ax(t) + Bu(t) \quad (2.1a)$$

$$y(t) = Cx(t) + Du(t) \quad (2.1b)$$

Where $t \in \mathbb{Z}_{\geq 0}$ is the time axis, $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the input and $y(t) \in \mathbb{R}^p$ is the output of the system.

The true state space dimension is n and system matrices $A \in M_n(\mathbb{R})$, $B \in M_{n \times m}(\mathbb{R})$, $C \in M_{p \times n}(\mathbb{R})$, $D \in M_{p \times m}(\mathbb{R})$.

In our data-based approach, those matrices (i.e. the system model) are generally unknown. However, an upper bound on state space dimension is given $N \geq n$. Now let's define the system trajectories.

Definition 2. A sequence of ordered triples $(u(t), x(t), y(t))_{t=0}^{\infty}$ is called an input-state-output trajectory if

$$\begin{bmatrix} x(t+1) \\ y(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$$

It can be deduced trivially that, from the linearity of (2.1), any linear combination of input-state-output trajectories is itself an input-state-output trajectory i.e. the set of all such trajectories spans a vector space over $\mathbb{R}$.

Additionally, from time invariance (2.1) we deduce that:

$$(u(t+\tau), x(t+\tau), y(t+\tau))_{t=0}^{\infty}$$

is an input-state-output trajectory for all $\tau \in \mathbb{N}$

Definition 3. A sequence of ordered pairs $(u(t), y(t))_{t=0}^{\infty}$ is called an input-output trajectory if there exists a sequence of states $x : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}^n$ such that $(u(t), x(t), y(t))_{t=0}^{\infty}$ is an input-state-output trajectory.

In practice we are limited by finite computer memory so we cannot work with infinite sequences of data. Motivated by that we state the next definition.

Definition 4. Let $i, j \in \mathbb{Z}_{\geq 0}$ such that $i \leq j$. Given an input-state-output trajectory $(u(t), x(t), y(t))_{t=0}^{\infty}$, the sequence $(u(t), x(t), y(t))_{t=i}^j$ is called a restricted input-state-output trajectory. Restricted input-output trajectories are defined in the same way.

2.1.2 The Concepts of Controllability and Observability

In this section the concepts of controllability and observability will be defined. Since, the LTI system (2.1) was defined classically it will be continued for these definitions. However, while the definitions of controllability/observability in classical and behavioural approach are almost equivalent, there is an assumption when equating those definitions with behaviour from input-output trajectories.

Definition 5. An LTI system (2.1) is controllable if for any initial state $x(0)$ and any desired state x_f , there exists a finite-length sequence of inputs

$$u(0), u(1), \dots, u(N-1)$$

such that the system reaches $x(N) = x_f$

In lay terms, it is telling that any state of the system can be reached in finite time by applying finitely many inputs.

Now the famous result for testing controllability will be stated. The pair (A, B) is the pair of matrices from (2.1) and n is the true state space dimension of the system.

Theorem 1. (Kalman controllability theorem) The pair (A, B) is controllable if and only if the controllability block matrix

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix}$$

is of full rank.

References

[1] Author Name. *Title of the Book or Article*. Publisher or Journal, year.

A Additional Material